

Electrostatics

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- Coulomb's Law
 - Force between two point charges: $F = k \frac{|q_1 q_2|}{r^2}$, where $k = \frac{1}{4\pi\epsilon_0}$
 - $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$
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- Electric Field
 - Electric field due to a point charge: $E = k \frac{|q|}{r^2}$
 - Electric field intensity: $\vec{E} = \frac{\vec{E}}{\epsilon_0}$
 - Electric dipole moment: $\vec{p} = q(2\vec{a})$
 - Torque on a dipole in uniform E-field: $\vec{\tau} = \vec{p} \times \vec{E}$
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- Electric Potential
 - Electric potential due to a point charge: $V = k \frac{q}{r}$
 - Potential energy of a dipole in E-field: $U = -\vec{p} \cdot \vec{E}$
 - Potential difference: $V_B - V_A = \frac{W_{AB}}{q_0}$
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- Gauss's Law
 - Electric flux: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$
 - For an infinite line charge: $E = \frac{\lambda}{2\pi\epsilon_0 r}$
 - For an infinite plane sheet: $E = \frac{\sigma}{2\epsilon_0}$
 - For a spherical shell (outside): $E = \frac{Q}{4\pi\epsilon_0 r^2}$
 - For a spherical shell (inside): $E = 0$

Capacitance

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- Capacitance
 - Definition: $C = \frac{Q}{V}$
 - Parallel plate capacitor: $C = \frac{\epsilon_0 A}{d}$
 - With dielectric: $C = \frac{K\epsilon_0 A}{d}$
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- Capacitors in Combination
 - Series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
 - Parallel: $C_{eq} = C_1 + C_2 + \dots$
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- Energy Stored
 - Energy stored in a capacitor: $U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$
 - Energy density: $u = \frac{1}{2} \epsilon_0 E^2$

Current Electricity

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- Electric Current
 - Current: $I = \frac{dQ}{dt}$
 - Drift velocity: $v_d = \frac{-eE\tau}{m}$
 - Relation between current and drift velocity: $I = nAev_d$
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- Ohm's Law and Resistance
 - Ohm's Law: $V = IR$
 - Resistance: $R = \rho \frac{L}{A}$
 - Resistivity: $\rho = \frac{m}{ne^2\tau}$
 - Temperature dependence of resistance: $R_T = R_0[1 + \alpha(T - T_0)]$
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- Resistances in Combination
 - Series: $R_{eq} = R_1 + R_2 + \dots$
 - Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
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- Kirchhoff's Laws
 - Junction Rule (KCL): $\sum I = 0$
 - Loop Rule (KVL): $\sum \Delta V = 0$
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- Power
 - Electric power: $P = VI = I^2 R = \frac{V^2}{R}$
 - Energy consumed: $E = P \times t$

Magnetism

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- Biot-Savart Law
 - Magnetic field due to a current element: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$

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- Magnetic field at center of circular loop: $B = \frac{\mu_0 I}{2R}$
- Magnetic field on axis of circular loop: $B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$
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- Ampere's Circuital Law
 - $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$
 - Magnetic field due to a straight infinite wire: $B = \frac{\mu_0 I}{2\pi r}$
 - Magnetic field inside a solenoid: $B = \mu_0 n I$
 - Magnetic field inside a toroid: $B = \frac{\mu_0 N I}{2\pi r}$
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- Force on Moving Charges and Current
 - Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$
 - Magnetic force on a charge: $\vec{F} = q(\vec{v} \times \vec{B})$
 - Magnetic force on a current-carrying conductor: $\vec{F} = I(\vec{L} \times \vec{B})$
 - Force between two parallel current-carrying wires: $F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} L$
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- Torque on a Current Loop
 - Magnetic dipole moment: $\vec{M} = IA\hat{n}$
 - Torque on a current loop: $\vec{\tau} = \vec{M} \times \vec{B}$
 - Moving coil galvanometer (deflection): $\alpha \propto I$

Electromagnetic Induction & AC

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- Faraday's Laws of EMI
 - Magnetic flux: $\Phi_B = \int \vec{B} \cdot d\vec{A}$
 - Induced EMF: $\epsilon = -\frac{d\Phi_B}{dt}$
 - Motional EMF: $\epsilon = BLv$
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- Self and Mutual Induction
 - Self-inductance: $\Phi = LI \Rightarrow \epsilon = -L \frac{dI}{dt}$
 - Mutual inductance: $\Phi_2 = MI_1 \Rightarrow \epsilon_2 = -M \frac{dI_1}{dt}$
 - Energy stored in an inductor: $U = \frac{1}{2} LI^2$
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- Alternating Current (AC)
 - RMS value: $I_{rms} = \frac{I_0}{\sqrt{2}}, V_{rms} = \frac{V_0}{\sqrt{2}}$
 - Inductive reactance: $X_L = \omega L = 2\pi f L$
 - Capacitive reactance: $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
 - Impedance (LCR series circuit): $Z = \sqrt{R^2 + (X_L - X_C)^2}$
 - Phase angle: $\tan \phi = \frac{X_L - X_C}{R}$
 - Power in AC circuit: $P = V_{rms} I_{rms} \cos \phi$ (Power factor $\cos \phi$)
 - Resonant frequency: $f_0 = \frac{1}{2\pi\sqrt{LC}}$
 - Transformer equation: $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

Electromagnetic Waves

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- Properties
 - Speed of EM waves in vacuum: $c = \frac{1}{\sqrt{\mu_0\epsilon_0}} = 3 \times 10^8 \text{ m/s}$
 - Speed of EM waves in medium: $v = \frac{1}{\sqrt{\mu\epsilon}}$
 - Relation between E and B fields: $E_0 = cB_0$
 - Displacement current: $I_D = \epsilon_0 \frac{d\Phi_E}{dt}$
 - Maxwell's Equations (integral form):
 - $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$ (Gauss's Law for Electricity)
 - $\oint \vec{B} \cdot d\vec{A} = 0$ (Gauss's Law for Magnetism)
 - $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$ (Faraday's Law)
 - $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$ (Ampere-Maxwell Law)

Optics

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- Reflection and Refraction
 - Mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
 - Magnification: $m = -\frac{v}{u} = \frac{h_i}{h_o}$
 - Refractive index: $n = \frac{c}{v}$
 - Snell's Law: $n_1 \sin i = n_2 \sin r$
 - Lens Maker's formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$
 - Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
 - Power of a lens: $P = \frac{1}{f}$ (in diopters, f in meters)
 - Total Internal Reflection (TIR): $\sin C = \frac{n_2}{n_1}$ (where $n_1 > n_2$)
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Wave Optics

- Huygens' Principle: Each point on a wavefront is a source of secondary wavelets.
- Young's Double Slit Experiment (YDSE):
 - Path difference: $\Delta x = d \sin \theta$
 - For bright fringes: $\Delta x = n\lambda$
 - For dark fringes: $\Delta x = (n + \frac{1}{2})\lambda$
 - Fringe width: $\beta = \frac{\lambda D}{d}$
- Diffraction (single slit):
 - Condition for minima: $a \sin \theta = n\lambda$
 - Condition for maxima: $a \sin \theta = (n + \frac{1}{2})\lambda$
 - Width of central maximum: $\frac{2\lambda D}{a}$
- Malus's Law: $I = I_0 \cos^2 \theta$ (Polarization)
- Brewster's Law: $\tan i_p = n$

Dual Nature of Radiation and Matter

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- Photoelectric Effect
 - Photon energy: $E = h\nu = \frac{hc}{\lambda}$
 - Einstein's photoelectric equation: $K_{max} = h\nu - \Phi_0 = h\nu - h\nu_0$
 - Work function: $\Phi_0 = h\nu_0$
 - Stopping potential: $eV_0 = K_{max}$
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- Matter Waves
 - de Broglie wavelength: $\lambda = \frac{h}{p} = \frac{h}{mv}$
 - For an electron accelerated through potential V: $\lambda = \frac{h}{\sqrt{2meV}}$

Atoms and Nuclei

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- Bohr's Model of Hydrogen Atom
 - Radius of n-th orbit: $r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} = 0.529 \frac{n^2}{Z} \text{ \AA}$
 - Energy of n-th orbit: $E_n = -\frac{m e^4}{8 \epsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2} \text{ eV}$
 - Wavelength of emitted photon: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$, where R is Rydberg constant
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- Nuclear Physics
 - Mass defect: $\Delta M = (Zm_p + Nm_n) - M_{nucleus}$
 - Binding energy: $E_b = \Delta M c^2$
 - Radioactive decay law: $N = N_0 e^{-\lambda t}$
 - Half-life: $T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$
 - Mean life: $\tau = \frac{1}{\lambda}$
 - Activity: $A = -\frac{dN}{dt} = \lambda N$

Semiconductor Electronics

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- Diodes
 - Forward bias: current flows
 - Reverse bias: very small current (leakage)
 - Ideal diode: acts as a short circuit in forward bias, open circuit in reverse bias.
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- Transistors (BJT)
 - Current relations: $I_E = I_B + I_C$
 - Current gain (common emitter): $\beta = \frac{I_C}{I_B}$
 - Alpha (common base): $\alpha = \frac{I_C}{I_E}$
 - Relation between alpha and beta: $\beta = \frac{\alpha}{1-\alpha}$
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- Logic Gates
 - Basic gates: AND, OR, NOT
 - Universal gates: NAND, NOR
 - Truth tables for all gates.